

PRASHANT RAJ

AI/ML Software Engineer | LLM Pipelines & RAG | Agentic AI | Python | B.Tech CSE (AI & ML)
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PROFESSIONAL SUMMARY

B.Tech Computer Science (AI & ML) student with hands-on experience building LLM-powered pipelines, RAG systems, and production-grade AI applications. Developed CodeSage AI — an end-to-end RAG pipeline integrating LangChain, Gemini LLM API, ChromaDB vector database, prompt engineering, and FastAPI to deliver accurate codebase intelligence. Solid Python foundations in OOP, data structures, APIs, and debugging. Brings a product-focused, first-principles mindset to designing scalable, reliable, and responsible AI systems. Seeking the Software Engineer AI/ML role at NucleusTeq to contribute to agentic AI development and LLM-powered data transformation.

TECHNICAL SKILLS

AI & LLM: Gemini API, LangChain (RAG + LLM orchestration), Prompt & Context Engineering, Embeddings, RAG Pipeline Design, Function Calling
Vector & Data Stores: ChromaDB, FAISS (conceptual), MongoDB, MySQL (SQL queries, joins, indexing, normalization), Supabase
Agentic & GenAI: RAG Architecture, LLM Pipelines, Hallucination Mitigation, Input/Output Validation, Responsible AI Practices
Python Stack: Python (OOP, data structures, scripting), FastAPI, Flask, Django, NumPy, Scikit-learn, TensorFlow, Keras
ML / Deep Learning: Convolutional Neural Networks (CNN), Medical Image Classification, Model Evaluation (Precision, Recall, F1)
Full-Stack & APIs: React.js, Node.js, Express.js, REST APIs, WebSockets, Socket.io, HTML5, CSS3, Bootstrap, Tailwind CSS
Tools & Practices: Git, GitHub, VS Code, Postman, Netlify, Modular Code Design, System Design Fundamentals
DSA: 190+ LeetCode problems (Java & Python) — DP, Graphs (BFS/DFS), Binary Search, Arrays, Hash Tables

PROJECTS

CodeSage AI – LLM-Powered Agentic Codebase Intelligence Engine | *Python, LangChain, Gemini API, ChromaDB, FastAPI, RAG, Prompt Engineering*
[\[GitHub\]](#) 2026 (In Progress)

- Architected an end-to-end LangChain-powered RAG pipeline: parses GitHub repos or ZIP uploads, generates vector embeddings, and indexes them in ChromaDB for semantic retrieval — enabling knowledge-driven query answering over real codebases.
- Engineered structured prompt templates and context injection strategies using LangChain's prompt and retrieval modules; grounded Gemini LLM responses in retrieved code to minimize hallucinations and ensure verifiable, project-specific answers.
- Exposed the pipeline via a FastAPI backend with input/output validation and modular agent-style components for code review, bug analysis, documentation generation, and test case creation — following responsible AI and clean code practices.

Malaria Detection – Deep Learning Medical Image Classifier | *Python, CNN, TensorFlow, Keras, NumPy, Scikit-learn, Flask* [\[GitHub\]](#) Jan – Mar 2026

- Trained a CNN on the NIH Malaria Cell Images Dataset to classify blood smear images (Parasitized / Uninfected); applied first-principles understanding of convolutional feature extraction and data augmentation pipelines.
- Evaluated model performance iteratively using precision, recall, and F1-score; optimized architecture to exceed baseline accuracy. Deployed via Flask API with an image upload interface for real-time inference.

UNICHAT – Real-Time Encrypted Chat Platform | *JavaScript, React, Node.js, Express.js, Socket.io, MongoDB, REST API* [\[Live\]](#) [\[GitHub\]](#) 2024 – 2025

- Built a full-stack real-time chat application with end-to-end encryption, WebSocket messaging (Socket.io), JWT authentication, and a modular OOP-based REST API backend — demonstrating system design and clean code fundamentals.
- Deployed live on Netlify with MongoDB for data persistence; implemented scalable API endpoints following microservice-style separation of concerns.

EDUCATION

B.Tech – Computer Science & Engineering (AI & ML)

Lakshmi Narain College of Technology & Science, RGPV, Bhopal

Key Subjects: AI & ML, Data Structures & Algorithms, OOP, DBMS & SQL, Computer Networks

CGPA: 8.05 / 10 | 2023 – 2027

Higher Secondary (Class XII) — RSSL College

72% | 2023

Secondary Education (Class X) — Regional Secondary School

63.6% | 2021

ACHIEVEMENTS & CERTIFICATIONS

- Built CodeSage AI — LangChain-powered RAG pipeline with Gemini API, ChromaDB, FastAPI; implementing responsible AI practices including input/output validation and hallucination mitigation.
- LeetCode: 190+ problems in Java & Python — Dynamic Programming, Graph Algorithms (BFS/DFS, Dijkstra), Binary Search (24), Arrays (88), Hash Tables (39).
- Certifications: Python Programming, Data Structures, HTML & Web Design, Cloud Computing Workshop, English IT Essentials.